3 GOOD HEALTH AND WELL-BEING

RYDEX

Two-Wheeler Accident Detection and Emergency Alert System

TEAM MEMBERS

Riya Bandekar
25P0320032

Rupali Arekar
25P0320035

Srinidhi Sawant
25P0320049

Sudipam Naik
25P0320050



Problem Statement:

Two-wheeler riders are vulnerable to road accidents, especially when they are travelling alone or in areas where immediate assistance may not be available. After an accident, injuries or unconsciousness may prevent riders from contacting emergency services or sharing their location with family members. This can delay the process of providing necessary assistance. From a Design Thinking perspective, there is a need for an affordable, user-friendly solution that addresses riders' safety concerns and emergency communication needs. From an IoT perspective, the challenge is to detect possible accidents, reduce false alerts, and communicate emergency information effectively. Therefore, the proposed system aims to integrate motion sensors, Bluetooth connectivity and a mobile application to detect possible accidents and notify selected emergency contacts with the rider's location after a confirmation countdown.

5W + 1H Analysis:

1. WHO — Who faces the problem?

Two-wheeler riders, daily commuters, college students, delivery workers and long-distance motorcyclists may face this problem. Family members and emergency responders may also be affected when accident information is delayed.

2. WHAT — What is the problem?

After an accident, a rider may be unconscious, injured or unable to use their phone. As a result, emergency contacts may not be informed quickly, and the rider's location may be difficult to communicate.

3. WHEN — When does the problem occur?

The problem occurs immediately after a road accident, particularly when the rider is alone, travelling at night, riding in an unfamiliar area or unable to call for assistance.

4. WHERE — Where does the problem occur?

It may occur on highways, rural roads, city streets, isolated routes and other locations where assistance may not be immediately available.

5. WHY — Why does this problem need to be solved?

Delayed communication can delay assistance. An injured rider may not be able to unlock their phone, make a call or share their location manually. An automated alert system could help reduce this communication delay.

6. HOW — How can the problem be addressed?

Develop a compact IoT device using an ESP32 and MPU6050 sensor to detect possible accidents. Connect it to an Android application through Bluetooth, provide a cancellation countdown, and use the phone's location and available communication services to alert selected emergency contacts.

Category	What	Why	Who	Where	When	How
Problem identification	Accident reporting is delayed, and sharing the rider's location is difficult.	Riders may be unable to report an accident or ask for help after it happens.	Two-wheeler riders and their emergency contacts.	Roads, highways and isolated locations.	During or immediately after an accident.	A sensor identifies a possible crash and sends an emergency report with location details.
IoT device and technology	A compact accident-detection device with motion sensors and Bluetooth.	To monitor motion and recognize possible accident events.	Riders using the device; family and emergency contacts.	Mounted on the two-wheeler or worn by the rider.	Before riding and during a ride.	ESP32, MPU6050, Bluetooth Low Energy, buzzer and protected power circuitry.
Target users	People who may benefit from automated accident alerts.	To support emergency communication when riders cannot contact others themselves.	Students, commuters, delivery workers and long-distance riders.	Cities, towns, rural areas and highways.	During daily travel and long journeys.	A rider-friendly connected Android application receives alerts and location information.
Smart detection features	Sensor-based detection of falls or impacts.	To identify potential accidents while reducing false alerts.	Riders and people relying on emergency notifications.	On the motorcycle, with status information in the app.	When unusual motion or a fall is detected.	Use acceleration, rotation, tilt and sustained-fall checks to assess a possible crash.
Emergency response	An accident alert with the rider's location.	To reduce reliance on the rider being able to call for help.	The rider, family members and emergency contacts.	Through the Android app and notification service.	After an accident is detected.	The app obtains the phone's location and shares an alert with selected contacts.
Safety and usability	Pre-ride connection check, status and alert-cancellation option.	To improve usability and reduce avoidable alerts.	Riders using the device and mobile app.	On the device and in the app.	Before riding, during connection loss, and after a detected incident.	Show readiness, battery, sensor and connection status; allow a short cancellation countdown.

Brainstorming:

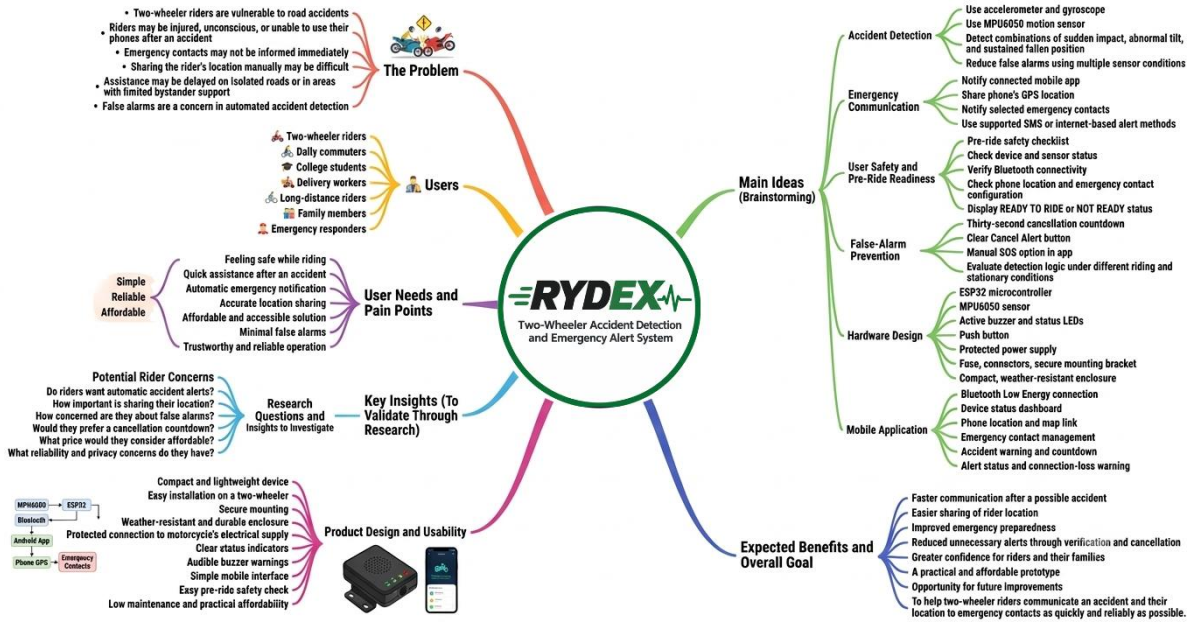
Brainstorming keywords and groups:

Accidents	Injuries	Unconsciousness
Delayed help	Rider alone	Remote roads
Emergency contacts	Location sharing	GPS
Bluetooth	Fall detection	Impact sensor
False alarms	30-second timer	Cancel alert
Manual SOS	Buzzer warning	Battery power
Weather protection	Affordable device	Easy installation
Pre-ride check	Mobile app	Family notification
Quick response	Reliable connection	

Group 1	Group 2	Group 3	Group 4	Group 5
Accident Challenges	Accident Detection	Emergency Communication	Smart Technology	Product Design and Usability
1.Accidents 2.Injuries 3.Unconsciousness 4.Rider alone 5.Remote roads 6.Delayed help	1.Fall detection 2.Impact sensor 3.Motion sensing 4.Tilt detection 5.False alarms	1.Emergency contacts 2.Location sharing 3.GPS 4.Family notification 5.Quick response 6.Manual SOS	1.Bluetooth 2.Mobile app 3.Pre-ride check 4.Reliable connection 5. 30-second timer 6.Cancel alert	1.Affordable device 2.Easy installation 3.Battery power 4.Weather protection 5.Buzzer warning

Mind map:

The mind map can organise the project into five branches: accident challenges, accident detection, emergency communication, smart technology, and product design and usability.



Proposed Solution

- Develop a compact IoT-based system that can detect a possible accident and notify selected emergency contacts after a short confirmation period.
- An ESP32 microcontroller to support device control and Bluetooth connectivity.
- An MPU6050 motion sensor to identify sudden movement or tilt patterns that may indicate a possible accident.
- Bluetooth connectivity between the device and an Android mobile application.
- A confirmation countdown that gives the rider an opportunity to cancel a false alert.
- The mobile application uses the phone's location and available communication services to notify selected emergency contacts.
- Design focus: keep the system affordable, easy to use, and practical for everyday riders.

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